



Unit 3 · Outcome 1 · Maintaining Biodiversity

Ranking a species by its risk of extinction

KK08

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



The Kestrel Ridge brief

00 The Kestrel Ridge brief The running scenario for this chapter — one reserve, several species, very different futures. Kestrel Ridge Reserve is a box-ironbark woodland in central Victoria with a creek wetland, grassy slopes and a rocky ridge. A field team has just finished a survey season. Five of the species they logged are on the desk of the reserve manager, who has to write a single funding bid. She can't say "they're all endangered" — the assessor will ask how endangered, and compared with what . ★ The job of a conservation category A conservation

One ladder, climbing toward extinction

01 One ladder, climbing toward extinction . The lower the rung, the higher the risk. Picture the categories as rungs on a ladder . The top rung is comfortable — the species is widespread and abundant. Each step down means a higher probability that the species disappears from the wild. The very bottom rung means it is already gone from the wild entirely. A species' category is just which rung it is standing on right now . The extinction-risk ladder Genetic → species → ecosystem... here: secure → lost live 3D ♡ . the Honeyeater marker sits on CR Each rung is

The six categories you must know

02 The six categories you must know VCAA wording: "extinct in the wild; critically endangered; endangered; vulnerable; near threatened; and least concern." Read these from the bottom of the ladder up — worst first — because that is the order conservation effort flows. Extinct (EX) sits below the VCAA list; it is the reference point the whole scale is measured against, so it's shown here in grey. EX Extinct beyond the list No reasonable doubt the last individual has died . There is no wild and no captive population anywhere. Extinction is permanent — ther

Status explorer — where the individuals are

03 Status explorer — where the individuals are Tap a category. Watch how many individuals survive, and whether any are left in the wild. The clearest way to feel the difference between the categories is to watch the population itself. In this scene the green plane is the wild (the species' natural range) and the glass box on the right is captivity — a zoo, sanctuary or breeding program. Each glowing dot is an individual. Step down the categories and two things change together: how many dots there are , and where they are allowed to be . Wild & ark popula

The "threatened" bracket & the traps around it

⚠ EXAM TRAPS

04 The "threatened" bracket & the traps around it Where most marks are won and lost. When a report, treaty or Act says a species is "threatened" , it is using a precise umbrella term that covers exactly three categories: critically endangered, endangered and vulnerable . That precision matters — a species that is near threatened is, by definition, not threatened yet, even though the name sounds like it should be. ! Trap 1 — "Near threatened" is not threatened The "near" is doing a lot of work. Near threatened means a species is approaching the threatened

A category is a snapshot, not a sentence

05 A category is a snapshot, not a sentence Species move up and down the ladder. Toggle decline vs recovery. A conservation category is not fixed . As threats grow, a species can be uplisted to a worse rung. As conservation works, it can be downlisted to a better one. The category is a snapshot of risk right now , and re-assessment is built into the system. Trajectory along the ladder Uplisting under threat · downlisting through recovery live 3D ↺ ▼ Decline (threats grow) ▲ Recovery (conservation) ↻ The real-world round trip The European bison was hunted

Analogies that make the ladder stick

06 Analogies that make the ladder stick The hospital triage analogy Walk into an emergency department and everyone there needs help — but a nurse triages them. Critically endangered is the patient being rushed straight to resuscitation; endangered is the urgent-but-stable case; vulnerable is serious and admitted; near threatened is "keep an eye on them in the waiting room"; least concern is the person who came in, got checked, and is fine to go home. Triage doesn't mean the healthy person can never get sick — it means scarce doctors go where the risk is

Status sorter

◇ ACTIVE RECALL

07 Status sorter Read each field scenario and pick the category it best matches. Instant feedback, running score. Scenario 1 of 6 — Pick the category that best fits the description above. Score 0 / 0

Decoding the command word

◆ COMMAND WORDS

08 Decoding the command word The verb in the question tells you the depth required. Tap a card. Identify / State 1 mk see what examiners expect → Define / Describe 1–2 mk see what examiners expect → Distinguish / Compare 2–4 mk see what examiners expect → Explain 2–4 mk see what examiners expect → Justify / Evaluate 3–6 mk see what examiners expect → Tap a command word above to reveal exactly what a marker is looking for.

P·E·L writing trainer

🔗 P·E·L WRITING

09 P·E·L writing trainer Point · Explain · Link — the skeleton of every strong extended response. P·E·L Build the answer one move at a time A 4-mark "explain" question, worked the way examiners reward. Q. The Helmeted Honeyeater at Kestrel Ridge is listed as critically endangered , while a population of the same birds is kept at a breeding sanctuary. Explain what "critically endangered" indicates about the wild birds, and how the sanctuary population relates to the "extinct in the wild" category. (4 marks) P Point — answer the question directly, first



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions

★ PRACTICE & SELF-MARK

10 Practice questions VCE-style. Open each to compare a weak and a strong response, read the marking guide, then self-mark. 19 Five questions · 19 marks. Try each one in your logbook before opening the responses. Award yourself marks honestly against each guide — the radial dock bottom-right tracks your total.

Recap & glossary

✦ RECAP / CHEAT SHEET

11 Recap & glossary ◆ The ladder Six VCAA categories rank a species by risk of extinction , from least concern (lowest risk) down to extinct in the wild (gone from the wild). ◆ "Threatened" = 3 Only critically endangered, endangered and vulnerable count as threatened. Near threatened and least concern do not. ◆ EX vs EW Extinct in the wild still has living individuals in captivity or naturalised — recovery is possible. Extinct means none survive anywhere. ◆ Risk, not count Categories reflect trajectory and probability , not just population size. KK09 cov

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.